
A SYSTEMATIC STUDY OF DESIGN CHOICES FOR MULTI-LABEL CHEST X-RAY CLASSIFICATION ON CHEXPert

Ma'moun Yosef

Department of Artificial Intelligence

University of Jordan

Amman, Jordan

mam0233493@ju.edu.jo

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ABSTRACT

Results on CheXpert are hard to compare because published setups differ in backbone, resolution, uncertain label handling, objective and ensembling all at once. This report fixes the pipeline and varies one factor at a time across roughly fifty runs that share the same splits, preprocessing and evaluation. Repeating configurations across seeds gives a noise floor of 0.001 mean AUROC on a large automatically labelled validation split and 0.004 to 0.017 on the two official radiologist sets, which is wider than many of the differences the study set out to measure. Read against that floor, input resolution up to 768×640 and pretraining on chest radiographs are the choices that reliably pay, and the architecture matters more than its size. The ImageNet label set, label smoothing and direct AUC optimization all fall inside the noise, while the handling of uncertain labels is worth up to 0.023 on one backbone. Ensembling pays when its members differ: seven runs of one backbone average to less than three different backbones do. The best single model reaches 0.9113 mean AUROC on the official test set and a plain average of three backbones reaches 0.9174, a paired bootstrap gain of +0.0061 with a 95% interval of $[+0.0005, +0.0118]$. The trained models and their configurations are released.

1 Introduction

CheXpert [1] is one of the standard benchmarks for multi-label chest radiograph classification, and results on it are difficult to compare. Published setups differ in backbone, input resolution, uncertain label handling, augmentation, objective and ensembling all at once, so a reported gain cannot be attributed to any one of those choices.

This report fixes the pipeline and varies one factor at a time. Roughly fifty runs share the same splits, preprocessing, augmentation and evaluation, so every difference between them comes from the design choice under test. Repeated seeds give a noise floor to read those differences against, which turns out to matter more than any single result here.

The contributions are a controlled comparison of the main design choices on CheXpert, a measurement of how much of the resulting spread is run to run noise, a final ensemble reaching 0.9174 mean AUROC on the official test set, and a public release of the trained models and their configurations.

2 Methods

2.1 Dataset and splits

All experiments use mainly CheXpert [1], restricted to frontal views. The targets are the five competition pathologies: Atelectasis, Cardiomegaly, Consolidation, Edema and Pleural Effusion.

The official training was therefore divided into a training part and a validation part. The division is 90/10 and split by patient, so no patient appears in both parts. The three official splits are already patient disjoint. Sizes are given in Table 1.

The official validation set holds only 200 studies, which is too small to give a stable signal, so the larger val19k split was carved out of the training data for that purpose. Later runs also scored val200 at every validation and used it to pick the best checkpoint. The official test500 set was never seen during training and is kept for final reporting only.

These exact splits are shared by every run, so any difference between runs comes from the design choice and not from the data. A big portion of the runs were trained at 384×320 from the downscaled CheXpert release, and some trained above that resolution from the original full resolution images.

Table 1: Splits used in every experiment. Counts are frontal images.

Split	Source	Images	Labels
train	official train, 90%	171,927	automatic labeler
val19k	official train, 10%	19,100	automatic labeler
valid200	official validation, 200 studies	202	radiologist consensus
test500	official test, 500 studies	518	radiologist consensus

2.2 Uncertain labels

CheXpert marks each pathology per study as positive, negative, uncertain or blank. Blank cells are treated as negative in every run. The uncertain cells are the open question, and five policies were compared:

- **U-Ones**: uncertain becomes positive.
- **U-Zeros**: uncertain becomes negative.
- **Certain only**: uncertain cells are masked out of the loss.
- **U-MultiClass**: the task gets a three way head over negative, positive and uncertain.
- **U-SelfTrained**: uncertain training cells take a soft probability predicted by a model that was itself trained on certain cells only.

Uncertain labels in val19k are always U-Ones for consistency. The policies are compared in Section 3.4.

2.3 Preprocessing and augmentation

Each image is resized to fit inside the target width and height with its aspect ratio preserved, then zero padded on the short side to reach the exact size. Nothing is stretched and nothing is cropped, so the anatomy keeps its true proportions. Resampling uses area interpolation. Images are normalized with the statistics of the data each backbone was pretrained on. The base geometry is 384×320 , and the higher resolution runs scale it up to 1600×1312 .

Contrast limited adaptive histogram equalization is a common preprocessing step for radiographs, so it was included as an ablation with a clip limit of 2.0 and an 8×8 tile grid, and it did not show a meaningful difference on ResNet-50. It is off everywhere else.

Augmentation is applied to the training split only, never to any validation or test split. Each transform is drawn independently with its own probability, so a given image may receive none, one or several of them. The settings are listed in Table 2 and are the same in every run. Horizontal flips are excluded, because mirroring a radiograph moves the heart to the wrong side.

Table 2: Training augmentation. It is applied to the training split only.

Transform	Range	Probability
Rotation	± 5 degrees	0.2
Translation	$\pm 5\%$ of width, height	0.2
Scale	0.95 to 1.05	0.2
Brightness	$\pm 10\%$	0.2
Contrast	$\pm 10\%$	0.2
Horizontal flip	not used	

2.4 Backbones and pretraining

Twelve backbones were fine-tuned, listed in Table 3. Parameter counts are the real counts of the built model, head included. All weights come from torchvision or timm [2] except the two chest X-ray models. Every backbone is fine tuned end to end, with no frozen layers.

Table 3: Backbones, parameter counts and the resolutions each was trained at.

Backbone	Pretraining	Params (M)	Resolutions
ResNet-50 [3]	ImageNet-1k	23.5	384×320
DenseNet-121 [4]	ImageNet-1k	7.0	384×320
DenseNet-201 [4]	ImageNet-1k	18.1	384×320
ConvNeXt-T [5]	ImageNet-1k, 22k	27.8	384×320
ConvNeXt-S [5]	ImageNet-1k	49.5	384×320
ConvNeXt-B [5]	ImageNet-22k	87.6	384×320 , 768×640 , 1600×1312
ConvNeXt-L [5]	ImageNet-22k	196.2	384×320 , 768×640
Swin-B [6]	ImageNet-1k, 22k	86.9	384×384
EfficientNetV2-M [7]	ImageNet-21k	52.9	384×320
CoAtNet-1 [8]	ImageNet-12k	41.0	384×320
ViT-B, Medical-MAE [9]	chest X-rays, MAE	86.0	384×320 to 768×640
ViT-B, RAD-DINO [10]	chest X-rays, DINOv2	86.6	784×644 , 1064×896

The ImageNet-22k, 21k and 12k entries were pretrained on the larger label set and then fine tuned on ImageNet-1k before being used here.

The last two rows are the only backbones that never saw natural images. Medical-MAE is a ViT-B trained with masked autoencoding [11] on roughly 0.5 million chest radiographs, and three of its released checkpoints were compared: the raw self-supervised backbone, the same backbone after fine tuning on ChestX-ray14 [12], and the one already fine tuned on CheXpert. RAD-DINO [10] is a ViT-B trained with DINOv2 [13] on chest radiographs, and it uses its own normalization statistics rather than the ImageNet ones. A separate run also pretrained ConvNeXt on ChestX-ray14 before moving to CheXpert, which tests the same idea with a convolutional backbone.

2.5 Training setup

Every run uses AdamW, a batch size of 64, mixed precision, and a cosine schedule with a one epoch linear warmup. The loss is binary cross entropy on the logits, computed over the five tasks at once and masked wherever a target is undefined. Training stops early when the monitored mean AUROC has not improved for a fixed number of validations.

The weight decay is 1×10^{-5} throughout. The runs that start from a backbone already pretrained on chest X-rays use a lower peak learning rate than the rest, because those weights already sit close to a good solution and a larger rate would only undo them. Those runs also use layer-wise learning rate decay of 0.75, so the rate is multiplied by 0.75 for each block further from the head, together with gradient clipping at a gradient norm of 1.0. No layer is frozen anywhere. The peak and minimum learning rate of every run is listed in Table 10.

Later runs add label smoothing of 0.05 to the targets, which keeps the model from driving its logits to saturation on labels that are themselves uncertain. An ablation was done with and without label smoothing, and label smoothing showed a very small gain, small enough to sit inside the seed noise reported in Section 3.5. It is kept in every later run.

The high resolution vision transformers are trained in two stages. Their position embeddings are tied to the patch grid, so changing the input size means resampling them, and going straight from the pretraining resolution to the target resolution would be too large a jump for that resampling. The first stage therefore trains at an intermediate resolution, and the second stage begins from its best checkpoint and trains at the full target resolution, resampling the position embeddings again.

One group of runs replaces the objective entirely. Instead of cross entropy they optimize an AUC margin loss [14], which approximates the headline metric directly, using the min-max optimizer that this loss requires. Mixed precision is disabled there and the model is warm started from a cross entropy model, both following the practice recommended for that loss. This changes the loss, the optimizer and the learning rate scale together, so it is not a single variable comparison and is reported as such.

2.6 Evaluation

The headline metric is the mean AUROC over the five pathologies, with the per class values and AUPRC also reported. Scoring is at the image level.

F1, precision, recall and specificity use one threshold per pathology chosen to maximize F1 on val19k. Those thresholds are then frozen and applied unchanged to val200 and test500.

During training the model is validated every few hundred optimizer steps on both val19k and val200. The checkpoint that is kept is the one with the best mean AUROC on the monitored set, and early stopping watches the same quantity.

The final ensampling results carry bootstrap confidence intervals over 10,000 resamples of the images.

3 Results

3.1 Architecture

Table 4 compares the backbones under the shared baseline setting, where everything except the architecture is held fixed.

On val19k the backbones span 0.7994 to 0.8064, a range of 0.0070 across architectures that differ by a factor of twenty eight in parameter count. On the radiologist sets the same models span 0.0245 on val200 and 0.0170 on test500, around three times the seed spread most configurations show in Section 3.5. The distance between the best and the weakest backbone here is therefore a real difference, even though neighbouring rows are not separable.

Nothing about capacity orders the results. ConvNeXt-L, at 196.2 million parameters, finishes below ConvNeXt-T at 27.8 million on both radiologist sets, and DenseNet-121 at 7.0 million lands within 0.0042 of ConvNeXt-L on test500. ResNet-50, the oldest design in the table, is fourth on test500 and ahead of both ConvNeXt-B and ConvNeXt-L.

The sets do not agree on a winner. ConvNeXt-L is first on val19k, while ConvNeXt-T leads both radiologist sets, at 0.8850 and 0.8896. The two Swin rows sit at the bottom of test500 despite being among the larger models. Most neighbouring rows differ by 0.002 to 0.005, which is too little to rank them, so only the distance between the top and the bottom of the table should be read as a difference.

Table 4: Backbones under the shared baseline setting. Resolution is 384×320 for every row except Swin-B, whose pretrained window size requires 384×384 .

Backbone	Pretraining	Pretrain res.	Params (M)	val19k	val200	test500
ResNet-50	IN-1k	224 ²	23.5	0.7994	0.8759	0.8836
DenseNet-121	IN-1k	224 ²	7.0	0.8028	0.8727	0.8750
DenseNet-201	IN-1k	224 ²	18.1	0.8029	0.8755	0.8750
EfficientNetV2-M	IN-21k	384 ²	52.9	0.8020	0.8766	0.8805
CoAtNet-1	IN-12k	224 ²	41.0	0.8026	0.8769	0.8824
Swin-B	IN-1k	384 ²	86.9	0.8022	0.8605	0.8726
Swin-B	IN-22k	384 ²	86.9	0.8036	0.8638	0.8761
ConvNeXt-T	IN-1k	224 ²	27.8	0.8043	0.8850	0.8896
ConvNeXt-S	IN-1k	224 ²	49.5	0.8056	0.8675	0.8789
ConvNeXt-B	IN-22k	384 ²	87.6	0.8062	0.8806	0.8775
ConvNeXt-L	IN-22k	384 ²	196.2	0.8064	0.8748	0.8792

3.2 Pretraining source

Table 5 holds the resolution fixed at the shared baseline and varies only what the backbone saw before CheXpert.

The ImageNet label set makes no difference that survives the noise. Swin-B gains 0.0033 on val200 and 0.0035 on test500 from the larger label set, both a fraction of the seed spread in Section 3.5. Pretraining on ten times more natural image labels does not carry into this task.

An intermediate stage on ChestX-ray14 behaves differently depending on how that stage is run. Copying the CheXpert schedule into it leaves ConvNeXt-B where it started, 0.8795 against 0.8775 on test500 and lower on val200. Running the same stage gently, at a tenth of the learning rate for four epochs, gives 0.8942 on test500, 0.0167 above the baseline,

and softening both stages gives 0.8899. ConvNeXt-L moves the same way but by less, gaining 0.0055 on val200 and 0.0019 on test500. Only the gentle ConvNeXt-B stage clears the noise floor. One explanation is that the harm comes from the intermediate stage overwriting the ImageNet features rather than from the extra data itself.

The backbones that were pretrained on chest radiographs from the start are the clearest result in this section. All three Medical-MAE checkpoints beat every ImageNet backbone in the table on test500, with the raw self-supervised checkpoint reaching 0.9046. Against the matched ConvNeXt-B, which uses the same uncertainty policy, label smoothing and resolution, the gain is 0.0061 on test500 and 0.0061 on val200. It costs nothing at inference, since this ViT-B is 86.0 million parameters against 87.6 million for ConvNeXt-B.

The three Medical-MAE checkpoints also show how little the ranking sets agree. The checkpoint already fine tuned on CheXpert is far ahead on val19k at 0.8161, a margin of 0.0076 over the next best, and yet it is last of the three on both radiologist sets. The raw checkpoint, which was never fine tuned on any labelled chest X-ray task, is the one that transfers best. Section 3.5 returns to this disagreement.

Table 5: Pretraining source at the shared 384×320 setting (384×384 for Swin-B). Gentle stage 1 means the ChestX-ray14 stage ran at 1×10^{-5} for 4 epochs instead of 1×10^{-4} for 10, with the CheXpert stage left untouched. The last four rows use the mixed uncertainty policy and label smoothing, so the ConvNeXt-B row is repeated there under those same settings as the matched reference.

Backbone	Pretraining	U-policy	LS	val19k	val200	test500
<i>ImageNet label set</i>						
ConvNeXt-T	IN-1k	U-Ones	no	0.8043	0.8850	0.8896
Swin-B	IN-1k	U-Ones	no	0.8022	0.8605	0.8726
Swin-B	IN-22k	U-Ones	no	0.8036	0.8638	0.8761
<i>ChestX-ray14 pretraining</i>						
ConvNeXt-B	IN-22k	U-Ones	no	0.8062	0.8806	0.8775
ConvNeXt-B	IN-22k + CXR14	U-Ones	no	0.8061	0.8759	0.8795
ConvNeXt-B	IN-22k + CXR14, gentle stage 1	U-Ones	no	0.8062	0.8919	0.8942
ConvNeXt-B	IN-22k + CXR14, gentle both stages	U-Ones	no	0.8070	0.8895	0.8899
ConvNeXt-L	IN-22k	U-Ones	no	0.8064	0.8748	0.8792
ConvNeXt-L	IN-22k + CXR14	U-Ones	no	0.8067	0.8803	0.8811
<i>Chest X-ray pretrained backbones</i>						
ViT-B	Medical-MAE	mixed	yes	0.8065	0.9030	0.9046
ViT-B	Medical-MAE + CXR14	mixed	yes	0.8085	0.8970	0.9020
ViT-B	Medical-MAE + CheXpert	mixed	yes	0.8161	0.8906	0.8994
ConvNeXt-B	IN-22k	mixed	yes	0.8038	0.8969	0.8985

3.3 Resolution

Table 6 raises the input resolution while holding the rest of the recipe fixed. This is the largest single lever in the study, and it is also the only one whose direction the radiologist sets agree on.

Going from 384×320 to 768×640 gains ConvNeXt-B 0.0052 on test500 and 0.0025 on val200. ConvNeXt-L gains 0.0195 and 0.0261 over the same step, although that pair also changes the uncertainty policy and label smoothing and so overstates the effect. Medical-MAE gains 0.0047 on test500 and 0.0094 on val200.

Above roughly 768×640 the gains stop. ConvNeXt-B at 1600×1312 scores the same 0.9037 on test500 as its own 768×640 run, and RAD-DINO loses 0.0054 going to 1064×896 . Only val200 keeps rising, which is a property of that set rather than of the models, as Section 3.5 shows.

Table 6: Input resolution. Within each block only the resolution changes, except the ConvNeXt-L pair, where the larger run also uses the mixed uncertainty policy and label smoothing. Time is wall clock on the GPU listed in Table 10.

Backbone	Resolution	Time	val19k	val200	test500
ConvNeXt-B	384×320	1h 31m	0.8038	0.8969	0.8985
ConvNeXt-B	768×640	4h 16m	0.8065	0.8994	0.9037
ConvNeXt-B	1600×1312	9h 46m	0.8030	0.9063	0.9037
ConvNeXt-L	384×320	2h 25m	0.8064	0.8748	0.8792
ConvNeXt-L	768×640	3h 18m	0.8062	0.9009	0.8987
ViT-B, Medical-MAE	384×320	1h 11m	0.8085	0.8970	0.9020
ViT-B, Medical-MAE	768×640	3h 12m	0.8076	0.9064	0.9067
ViT-B, RAD-DINO	784×644	5h 15m	0.8043	0.9016	0.9100
ViT-B, RAD-DINO	1064×896	6h 14m	0.8021	0.9000	0.9046

3.4 Training choices

Table 7 covers what to do with the uncertain labels, what to optimize, and whether to smooth the targets. These runs sit on three different backbones, so only comparisons inside a block mean anything.

On ResNet-50, U-Zeros loses 0.0171 on val200 and 0.0040 on test500, so mapping uncertain cells to negative is the weaker choice there. On DenseNet-121 both alternatives to plain U-Ones help: sending Consolidation’s uncertain cells to negative gains 0.0176 on test500, and the full per task scheme gains 0.0228. Reading uncertain cells as positive, the simplest policy of all, is not the best on either backbone but is never far off.

On ConvNeXt-B the picture is clearer. Every variant that stops training Consolidation toward the labeler’s convention beats the U-Ones baseline on test500: masking uncertain cells out of the loss gives 0.8963, self-training all five tasks gives 0.8970, and self-training Consolidation alone gives 0.9002, against 0.8775 for the baseline. That last gain of 0.0227 is the largest single effect in this table. The curriculum variant, which shows the uncertain rows first, gives most of it back at 0.8848.

Label smoothing of 0.05 changes almost nothing. Against the run it was added to, it moves test500 by -0.0017 and val200 by -0.0002 , both well inside the noise. It is kept in the later runs.

Replacing cross entropy with the AUC margin loss also fails to pay. Warm started from a cross entropy model it reaches 0.8836 on test500, above the U-Ones baseline it warm started from but below several plain cross entropy runs. When it is trained alone, without the cross entropy stage, it falls to 0.7548 on val19k and 0.8539 on test500, the worst result in the study. Optimizing the metric directly is not a shortcut to it.

Table 7: Uncertainty policy, objective and label smoothing. Only comparisons inside one backbone block are meaningful. Mixed means the per task scheme of Section 2.2, with three way heads on Cardiomegaly and Pleural Effusion. val19k is scored under each run’s own policy, so it is not comparable across rows; the radiologist sets have no uncertain labels and are.

Run	Uncertainty and objective	val19k	val200	test500
<i>ResNet-50</i>				
baseline	U-Ones	0.7994	0.8759	0.8836
u_zeros	U-Zeros	0.7990	0.8588	0.8796
<i>DenseNet-121</i>				
baseline	U-Ones	0.8028	0.8727	0.8750
cons_zeros	U-Ones, U-Zeros on Consolidation	0.8085	0.8834	0.8926
u_mixed	mixed, U-Zeros on Consolidation	0.8159	0.8820	0.8978
<i>ConvNeXt-B</i>				
baseline	U-Ones	0.8062	0.8806	0.8775
certain_only	uncertain masked out of the loss	0.7983	0.8808	0.8963
u_selftrain_all	U-SelfTrained on all five tasks	0.7983	0.8856	0.8970
u_selftrain_consol_only	U-Ones, U-SelfTrained on Consolidation	0.8038	0.8971	0.9002
u_selftrain	mixed, U-SelfTrained on Consolidation	0.8220	0.8977	0.8977
curriculum	as consol_only, uncertain rows first	0.7984	0.8812	0.8848
final_stage1	as consol_only, plus label smoothing	0.8038	0.8969	0.8985
aucm	U-Ones, AUC-M after cross entropy	0.8069	0.8825	0.8836
aucm_no_ce	mixed, AUC-M alone	0.7548	0.8471	0.8539

3.5 How much of this is noise

Every number quoted so far should be read against the size of the differences that appear when nothing changes at all. Table 8 rebuilds the same configuration under different random seeds.

The noise floor depends on the set. On val19k, with 19,100 images, three ConvNeXt-B replicates span 0.0010 and three DenseNet-121 replicates span 0.0005. On the radiologist sets the same replicates span 0.0053 to 0.0156 on val200 and 0.0036 to 0.0170 on test500. So the DenseNet-121 block shows that a single unlucky seed can move test500 by 0.0170 on its own.

The three sets also measure different things, which is why they keep disagreeing. val19k is large and therefore stable, but its labels come from the automatic labeler and its uncertain cells are always scored as positive, so it rewards models that reproduce the labeler. This shows up directly: the Medical-MAE checkpoint already fine tuned on CheXpert leads val19k by 0.0076 and is mid table on both radiologist sets, while the run that stopped training toward the labeler convention altogether, the one that masks uncertain cells out of the loss, gives up 0.0079 on val19k and gains 0.0188 on test500.

val200 and test500 carry radiologist consensus labels, which is what the task actually asks for, but at 202 and 518 images they are too small to rank models that sit within 0.015 of each other. Their own disagreement is the proof: the ordering of the twelve backbones does not survive the move from one set to the other. A useful thought for the rest of this report is to read differences below roughly 0.015 on test500 as ties, and to take a lead on any single set as a hint rather than a settled ranking.

Table 8: Seed replicates. Every run inside a block is identical except for the random seed. Spread is the largest value minus the smallest.

Run	Seed	val19k	val200	test500
<i>ConvNeXt-B</i> , 384×320				
final_stage1	42	0.8038	0.8969	0.8985
seed7	7	0.8034	0.8916	0.8949
seed1337	1337	0.8028	0.8924	0.8967
<i>spread</i>		0.0010	0.0053	0.0036
<i>DenseNet-121</i> , 384×320				
densenet121	42	0.8028	0.8727	0.8750
seed7	7	0.8023	0.8740	0.8804
seed123	123	–	0.8584	0.8634
<i>spread</i>		0.0005	0.0156	0.0170
<i>ViT-B Medical-MAE</i> , 768×640				
nih_B_768_s2	42	0.8076	0.9064	0.9067
seed7	7	0.8090	0.9045	0.9077
seed1337	1337	0.8053	0.9105	0.9112
<i>spread</i>		0.0037	0.0059	0.0046

3.6 Ensembling and final result

Table 9 collects the ensembles. What decides their quality is not how many members they hold but how different those members are.

Averaging five checkpoints of one ConvNeXt-B reaches 0.9076, and widening that to seven ConvNeXt-B runs, including separate seeds, reaches 0.9095. Both sit below the best single model, which scores 0.9113 on its own. Replacing those seven near copies with three genuinely different backbones, ConvNeXt-B at 1600×1312 , Medical-MAE at 768×640 and RAD-DINO at 784×644 , lifts a plain probability average to **0.9174**, the best result in the study. Fitting per class weights on val200 does not improve on that plain average.

Over 10,000 paired resamples of the 518 test500 images, that ensemble scores 0.9174 with a 95% interval of [0.9040, 0.9303], and its gain over the strongest single member is +0.0061 with a 95% interval of [+0.0005, +0.0118]. Per class the ensemble reaches 0.9712 on Pleural Effusion, 0.9269 on Edema, 0.9138 on Cardiomegaly, 0.9127 on Consolidation and 0.8623 on Atelectasis.

Table 9: Ensembles on test500. Weights, when used, are per class and fitted on val200. The single model row is the strongest member on its own.

Members	Blend	Weights	test500
Medical-MAE ViT-B 768×640 alone	–	–	0.9113
5 ConvNeXt-B checkpoints	probability average	no	0.9076
7 ConvNeXt-B runs	probability average	no	0.9095
3 backbones	probability average	no	0.9174
8 backbones	logit average	yes	0.9159

4 Discussion

Across roughly fifty runs, a few choices moved the metric and most of the rest did not.

Resolution moved it, up to a point. Going from 384×320 to 768×640 was the largest single gain in the study, and going further gave almost nothing back. Domain pretraining moved it: backbones pretrained on chest radiographs beat every ImageNet backbone at equal size and equal resolution, which is the one place where the choice of starting weights clearly paid. Ensembling moved it, but only when the members were genuinely different from one another.

Architecture moved it too, but not in the way parameter counts suggest. The eleven backbones span 0.0170 on test500 and 0.0245 on val200, around three times the seed spread that most configurations show, so the difference between a good backbone and a weak one is real. What does not follow is capacity: ConvNeXt-T at 27.8 million parameters

finishes above ConvNeXt-L at 196.2 million on both radiologist sets, and on test500 even ResNet-50 lands ahead of ConvNeXt-B and ConvNeXt-L. Picking a good family matters, and paying for a larger member of that family does not.

What did not move the metric is a shorter list than expected. Enlarging the ImageNet label set from 1k to 22k changed nothing that survives the noise, on either backbone where both were available. That is a statement about the label set alone: every backbone here starts from pretrained weights and none was trained from scratch, so this study says nothing about the value of ImageNet pretraining itself, which is almost certainly large. The uncertainty policy, which the CheXpert literature has spent considerable effort on, mattered much less than the seed. Label smoothing did nothing measurable. Optimizing AUROC directly through an AUC margin loss did not beat cross entropy, and without a cross entropy stage it collapsed.

5 Limitations

Everything here is one dataset. There is no external validation on MIMIC-CXR, ChestX-ray14 as a target, or any prospective set, so none of these conclusions are known to transfer.

The final ensemble is selected and weighted on val200, which holds 202 images and, for Consolidation, 32 positives. The bootstrap interval on test500 accounts for sampling the test images but not for that selection, so any headline figure on test500 could be optimistic, and this applies to any published number on that set. The per class weights should be read as one draw from a noisy procedure rather than as a finding about which model is good at which pathology.

Not every ablation shares one backbone. The uncertainty policies are spread across ResNet-50, DenseNet-121 and ConvNeXt-B, and the ChestX-ray14 pretraining runs vary the schedule alongside the extra stage, so a few comparisons in Section 3.4 and Section 3.2 carry more than one variable. Only three configurations were repeated across seeds, so the noise floor is estimated from a small sample and may be understated for the configurations that were never repeated.

6 Conclusion

Under one fixed pipeline on CheXpert, input resolution, chest X-ray pretraining and the handling of uncertain labels were the design choices that paid, ensembling paid when its members were diverse, and the architecture mattered while its size did not as much. The remaining choices, including the ImageNet label set, label smoothing and the objective, were within the noise of rerunning the same configuration with a different seed. The best single model reached 0.9113 mean AUROC on the official test set and a plain average of three different backbones reached 0.9174, with a paired bootstrap interval of $[+0.0005, +0.0118]$ on the difference. The wider point is that the official radiologist sets are small enough that model rankings drawn from them are unstable, and reported gains below roughly 0.015 should be treated as ties.

References

- [1] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghighi, Robyn Ball, Katie Shpanskaya, Jayne Seekins, David A. Mong, Safwan S. Halabi, Jesse K. Sandberg, Ricky Jones, David B. Larson, Curtis P. Langlotz, Bhavik N. Patel, Matthew P. Lungren, and Andrew Y. Ng. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 590–597, 2019.
- [2] Ross Wightman. Pytorch image models. <https://github.com/huggingface/pytorch-image-models>, 2019.
- [3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.
- [4] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017.
- [5] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11976–11986, 2022.
- [6] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 10012–10022, 2021.

- [7] Mingxing Tan and Quoc V. Le. Efficientnetv2: Smaller models and faster training. In *Proceedings of the 38th International Conference on Machine Learning*, pages 10096–10106, 2021.
- [8] Zihang Dai, Hanxiao Liu, Quoc V. Le, and Mingxing Tan. Coatnet: Marrying convolution and attention for all data sizes. In *Advances in Neural Information Processing Systems*, volume 34, pages 3965–3977, 2021.
- [9] Junfei Xiao, Yutong Bai, Alan Yuille, and Zongwei Zhou. Delving into masked autoencoders for multi-label thorax disease classification. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 3588–3600, 2023.
- [10] Fernando Pérez-García, Harshita Sharma, Sam Bond-Taylor, Kenza Bouzid, Valentina Salvatelli, Maximilian Ilse, Shruthi Bannur, Daniel C. Castro, Anton Schwaighofer, Matthew P. Lungren, Maria Wetscherek, Noel Codella, Stephanie L. Hyland, Javier Alvarez-Valle, and Ozan Oktay. RAD-DINO: Exploring scalable medical image encoders beyond text supervision, 2024.
- [11] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16000–16009, 2022.
- [12] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2097–2106, 2017.
- [13] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.
- [14] Zhuoning Yuan, Yan Yan, Milan Sonka, and Tianbao Yang. Large-scale robust deep AUC maximization: A new surrogate loss and empirical studies on medical image classification. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3040–3049, 2021.

A Run configurations

Epochs are the configured budget, with the total number of optimizer steps in parentheses. Training stops early, so the selected checkpoint is usually well before the end. Batch size is 64 everywhere except `convnext_base_22k_1600x1312`, which used 32.

Table 10: Configuration and results of every run reported in this study.

Run	Pretraining	Resolution	LR peak	LR min	Epochs (steps)	GPU	U-policy	LS	Time	val19k	val200	test500
<i>Architecture</i>												
resnet50_without_clahe	IN-1k	384×320	1×10^{-4}	1×10^{-6}	15 (40,305)	L4	U-Ones	no	2h 08m	0.7994	0.8759	0.8836
densenet121	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	2h 15m	0.8028	0.8727	0.8750
densenet201	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	3h 06m	0.8029	0.8755	0.8750
convnext_tiny	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	1h 58m	0.8043	0.8850	0.8896
convnext_tiny_22k	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	2h 07m	0.8045	0.8788	0.8835
convnext_small	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	3h 17m	0.8056	0.8675	0.8789
convnext_base_22k	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	no	1h 17m	0.8062	0.8806	0.8775
convnext_large_22k	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	no	2h 25m	0.8064	0.8748	0.8792
swin_base	IN-1k	384×384	1×10^{-4}	1×10^{-6}	10 (26,870)	A100-80GB	U-Ones	no	2h 17m	0.8022	0.8605	0.8726
swin_base_22k	IN-22k	384×384	1×10^{-4}	1×10^{-6}	10 (26,870)	A100-80GB	U-Ones	no	2h 29m	0.8036	0.8638	0.8761
efficientnetv2_m	IN-21k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	yes	1h 18m	0.8020	0.8766	0.8805
coatnet_1_12k	IN-12k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	yes	1h 54m	0.8026	0.8769	0.8824
<i>Preprocessing</i>												
resnet50_with_clahe	IN-1k	384×320	1×10^{-4}	1×10^{-6}	15 (40,305)	L4	U-Ones	no	3h 39m	0.7973	0.8664	0.8732
<i>Uncertain labels</i>												
resnet50_u_zeros	IN-1k	384×320	1×10^{-4}	1×10^{-6}	15 (40,305)	L4	U-Zeros	no	2h 38m	0.7990	0.8588	0.8796
densenet121_cons_zeros	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	2h 13m	0.8085	0.8834	0.8926
densenet121_u_mixed	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	mixed	no	2h 39m	0.8159	0.8820	0.8978
convnext_base_22k_certain_only	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	no	1h 28m	0.7983	0.8808	0.8963
convnext_base_22k_u_selftrain	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	no	1h 47m	0.8220	0.8977	0.8977
convnext_base_22k_u_selftrain_all	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	mixed	no	3h 37m	0.7983	0.8856	0.8970
convnext_base_22k_u_selftrain_consol_only	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	no	1h 21m	0.8038	0.8971	0.9002
convnext_base_22k_curriculum	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (16,818)	A100	mixed	no	58m 11s	0.7984	0.8812	0.8848
<i>Label smoothing</i>												
convnext_base_22k_final_stage1	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	yes	1h 31m	0.8038	0.8969	0.8985
<i>Objective</i>												
convnext_base_22k_aucm	IN-22k	384×320	0.075	7.5×10^{-4}	6 (16,122)	A100	U-Ones	no	1h 15m	0.8069	0.8825	0.8836
convnext_base_22k_aucm_no_ce	IN-22k	384×320	0.075	7.5×10^{-4}	6 (16,122)	A100	mixed	yes	1h 40m	0.7548	0.8471	0.8539
<i>Domain pretraining</i>												
medmae_vitb_raw	MAE	384×320	5×10^{-5}	5×10^{-7}	6 (16,122)	A100	mixed	yes	1h 03m	0.8065	0.9030	0.9046
medmae_vitb_nih	MAE+CXRI4	384×320	5×10^{-5}	5×10^{-7}	6 (16,122)	A100	mixed	yes	1h 11m	0.8085	0.8970	0.9020
medmae_vitb_chexpert	MAE+CheXpert	384×320	1×10^{-5}	1×10^{-7}	6 (16,122)	A100	mixed	yes	1h 04m	0.8161	0.8906	0.8994
convnext_base_22k_cxr14_pretrain	IN-22k+CXRI4	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	no	1h 33m	0.8061	0.8759	0.8795
convnext_base_22k_cxr14_pretrain_lowlr	IN-22k+CXRI4	384×320	1×10^{-5}	1×10^{-7}	4 (26,870)	A100	U-Ones	no	1h 12m	0.8062	0.8919	0.8942
convnext_base_22k_cxr14_pretrain_lowlr_all	IN-22k+CXRI4	384×320	1×10^{-5}	1×10^{-7}	4 (16,122)	A100	U-Ones	no	1h 35m	0.8070	0.8895	0.8899
convnext_large_22k_cxr14_pretrain	IN-22k+CXRI4	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	U-Ones	no	1h 30m	0.8067	0.8803	0.8811
<i>Resolution</i>												
convnext_base_22k_768x640	IN-22k	768×640	1×10^{-4}	1×10^{-6}	10 (26,870)	A100-80GB	mixed	yes	4h 16m	0.8065	0.8994	0.9037
convnext_base_22k_1600x1312	IN-22k	1600×1312	1×10^{-4}	1×10^{-6}	10 (26,870)	B200	mixed	yes	9h 46m	0.8030	0.9063	0.9037
convnext_large_22k_768x640	IN-22k	768×640	1×10^{-4}	1×10^{-6}	10 (26,870)	H200	mixed	yes	3h 18m	0.8062	0.9009	0.8987
medmae_vitb_nih_B_768_s2	MAE+CXRI4	768×640	5×10^{-5}	5×10^{-7}	8 (21,496)	A100-80GB	mixed	yes	3h 12m	0.8076	0.9064	0.9067
rad_dino_vitB_768	DINOv2	784×644	3×10^{-5}	3×10^{-7}	5 (13,435)	A100-80GB	mixed	yes	5h 15m	0.8044	0.9016	0.9100
rad_dino_vitB_1064x896	DINOv2	1064×896	6×10^{-6}	1×10^{-7}	6 (16,122)	H200	mixed	yes	6h 14m	0.8021	0.9000	0.9046

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Table 10 continued from the previous page

Run	Pretraining	Resolution	LR peak	LR min	Epochs (steps)	GPU	U-policy	LS	Time	val19k	val200	test500
<i>Seed replicates</i>												
convnext_base_22k_seed7	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	yes	51m 18s	0.8034	0.8916	0.8949
convnext_base_22k_seed1337	IN-22k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	A100	mixed	yes	54m 20s	0.8028	0.8924	0.8967
densenet121_seed7	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	2h 28m	0.8023	0.8740	0.8804
densenet121_seed123	IN-1k	384×320	1×10^{-4}	1×10^{-6}	10 (26,870)	L4	U-Ones	no	2h 41m	0.8023	0.8584	0.8634
medmae_vitb_nih_B_448_s1_seed7	MAE+CXR14	448×384	5×10^{-5}	1×10^{-5}	1 (2,687)	A100	mixed	yes	39m 24s	0.7960	0.8972	0.9080
medmae_vitb_nih_B_448_s1_seed1337	MAE+CXR14	448×384	5×10^{-5}	1×10^{-5}	1 (2,687)	A100	mixed	yes	33m 06s	0.8019	0.9052	0.9088
medmae_vitb_nih_B_768_s2_seed7	MAE+CXR14	768×640	5×10^{-5}	5×10^{-7}	8 (21,496)	A100-80GB	mixed	yes	2h 22m	0.8090	0.9045	0.9077
medmae_vitb_nih_B_768_s2_seed1337	MAE+CXR14	768×640	5×10^{-5}	5×10^{-7}	8 (21,496)	A100-80GB	mixed	yes	2h 06m	0.8053	0.9105	0.9112

B Per-disease runs

These runs train a single pathology rather than all five, so their scores are single class AUROC and are not comparable to the mean AUROC reported elsewhere. The first group trains one ConvNeXt-B per pathology with cross entropy. The second group replaces the objective with the AUC margin loss, warm started from a shared cross entropy backbone.

Table 11: Single pathology runs. Scores are single class AUROC.

Run	Pathology	Objective	val19k	val200	test500
convnext_base_22k_atelectasis_only	Atelectasis	BCE	0.7243	0.8500	0.8431
convnext_base_22k_cardiomegaly_only	Cardiomegaly	BCE	0.8716	0.8277	0.8850
convnext_base_22k_consolidation_only	Consolidation	BCE	0.6800	0.8947	0.9155
convnext_base_22k_edema_only	Edema	BCE	0.8517	0.9298	0.9153
convnext_base_22k_pleural_effusion_only	Pleural Effusion	BCE	0.8782	0.9299	0.9495
convnext_base_22k_final_stage2_atelectasis	Atelectasis	AUC-M	0.7233	0.8557	0.8363
convnext_base_22k_final_stage2_cardiomegaly	Cardiomegaly	AUC-M	0.8712	0.8319	0.8949
convnext_base_22k_final_stage2_consolidation	Consolidation	AUC-M	0.6820	0.9175	0.8851
convnext_base_22k_final_stage2_edema	Edema	AUC-M	0.8485	0.9454	0.9232
convnext_base_22k_final_stage2_pleural_effusion	Pleural Effusion	AUC-M	0.8767	0.9324	0.9557